

Projeto 02

Unidade Curricular de Segurança

Artur Meireles de Sousa Falcão

a2024130485@isec.pt

2024130485

Data de entrega: 04/03/2026

Tempo de realização: ~3h + módulos Cisco

4.

b) Interfaces Ethernet0/0 e Ethernet0/1 encontram-se no estado “up” e com os endereços de IP corretamente configurados.

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1-0485
R1-0485(config)#EXIT
R1-0485#
*Mar 3 12:54:12.440: %SYS-5-CONFIG_I: Configured from console by console
R1-0485#copy running-config startup-config
Destination filename [startup-config]? startup-config
Building configuration...
[OK]
R1-0485#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	192.168.1.50	YES	NVRAM	up	up
Ethernet0/1	192.168.2.51	YES	NVRAM	up	up
Ethernet0/2	unassigned	YES	NVRAM	administratively down	down
Ethernet0/3	unassigned	YES	NVRAM	administratively down	down
Ethernet1/0	unassigned	YES	NVRAM	administratively down	down
Ethernet1/1	unassigned	YES	NVRAM	administratively down	down
Ethernet1/2	unassigned	YES	NVRAM	administratively down	down
Ethernet1/3	unassigned	YES	NVRAM	administratively down	down
Serial2/0	unassigned	YES	NVRAM	administratively down	down
Serial2/1	unassigned	YES	NVRAM	administratively down	down
Serial2/2	unassigned	YES	NVRAM	administratively down	down
Serial2/3	unassigned	YES	NVRAM	administratively down	down
Serial3/0	unassigned	YES	NVRAM	administratively down	down
Serial3/1	unassigned	YES	NVRAM	administratively down	down
Serial3/2	unassigned	YES	NVRAM	administratively down	down
Serial3/3	unassigned	YES	NVRAM	administratively down	down

e) Tentativa de acesso ao R1 através do comando telnet deu erro pois as linhas VTY ainda não foram configuradas para aceitar ligações telnet.

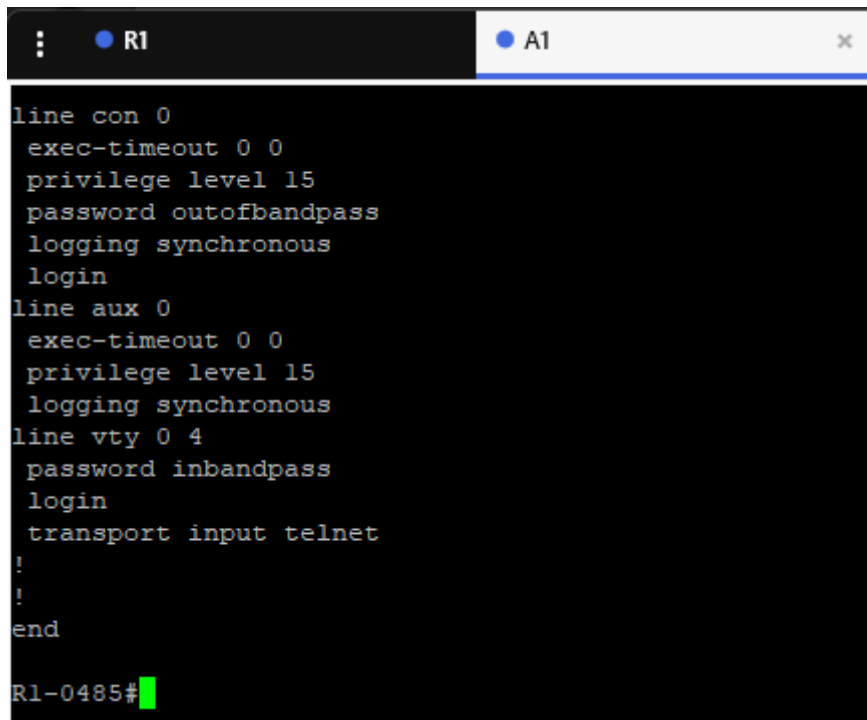
```
A1 console is now available... Press RETURN to get started.
root@A1:~# ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.85 netmask 255.255.255.0 broadcast 0.0.0.0
    inet6 fe80::42:40ff:fe47:af00 prefixlen 64 scopeid 0x20<link>
    ether 06:23:04:85:aa:01 txqueuelen 1000 (Ethernet)
    RX packets 29 bytes 4852 (4.7 KiB)
    RX errors 0 dropped 2 overruns 0 frame 0
    TX packets 12 bytes 936 (936.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@A1:~# telnet 192.168.1.50
Trying 192.168.1.50...
telnet: Unable to connect to remote host: Connection refused
```

g) Sim, a ligação remota já foi possível.

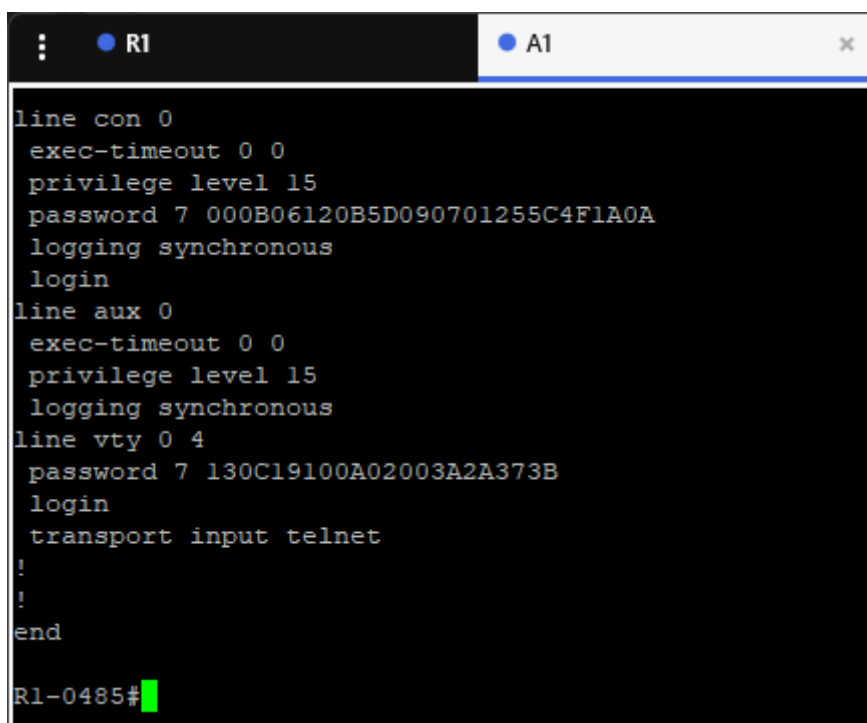
h) Sim, mas agora é exigida uma password para entrar no modo de configuração através do comando enable.

i) Através do comando “show running-config” verificamos que as passwords estão guardadas em “cleartext”.



```
line con 0
  exec-timeout 0 0
  privilege level 15
  password outofbandpass
  logging synchronous
  login
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  password inbandpass
  login
  transport input telnet
!
!
end
R1-0485#
```

j) Com a ativação do serviço de encriptação através do comando “service password-encryption”, as passwords passaram a estar ocultas na configuração.

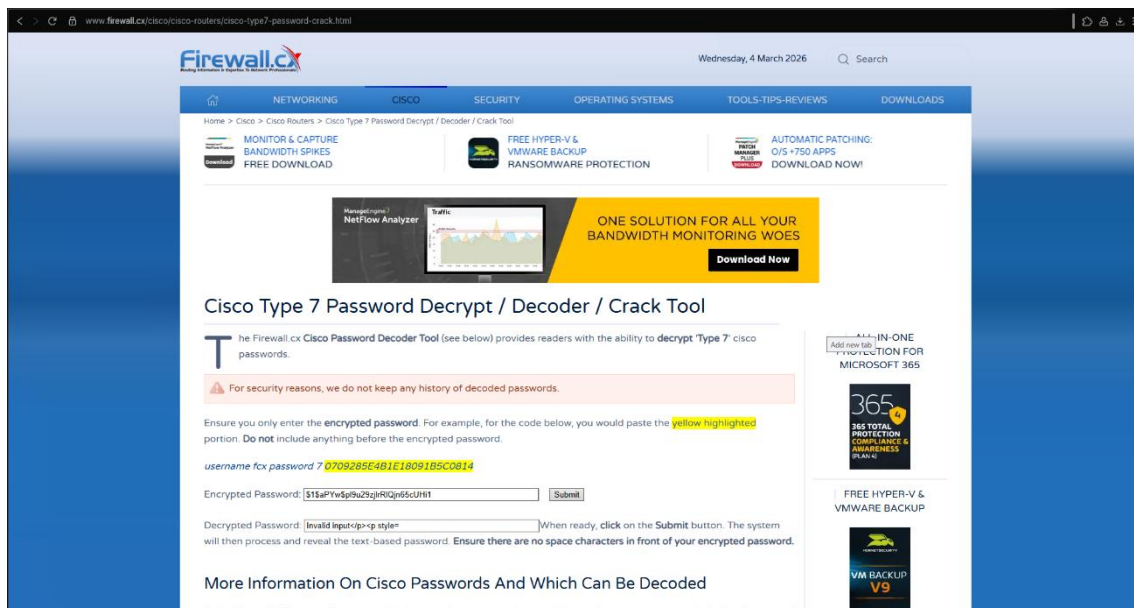


```
line con 0
  exec-timeout 0 0
  privilege level 15
  password 7 000B06120B5D090701255C4F1A0A
  logging synchronous
  login
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  password 7 130C19100A02003A2A373B
  login
  transport input telnet
!
!
end
R1-0485#
```

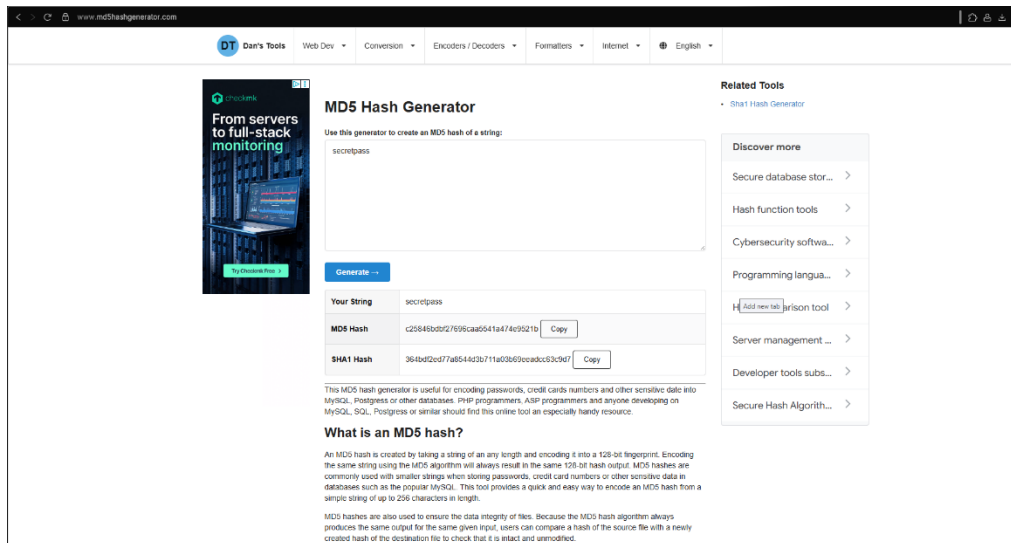
k) Utilizando o site “Firewall.cx”, foi possível reverter imediatamente o encriptamento da password.



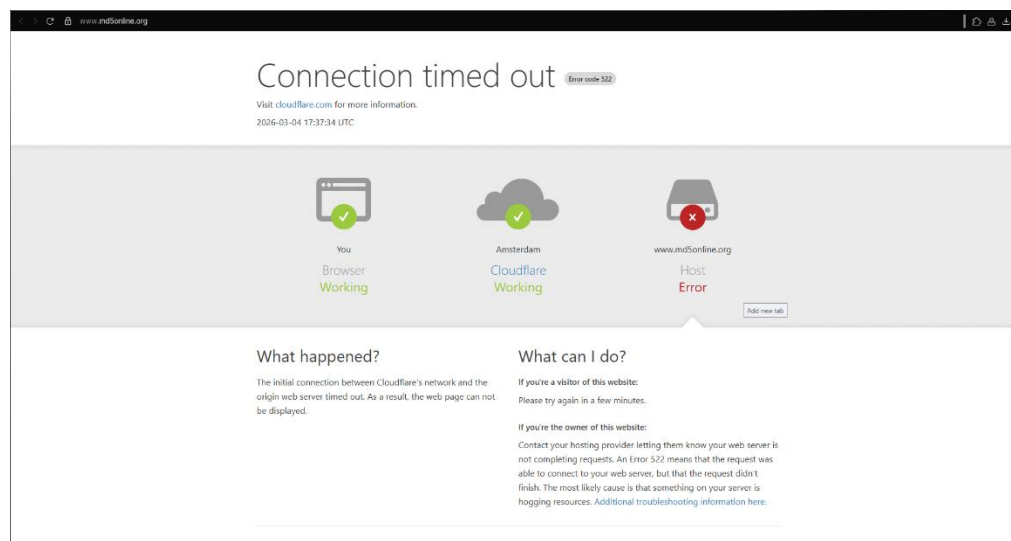
l) Ao repetir o procedimento com a password do “privileged EXEC mode” não foi possível reverter a encriptação.



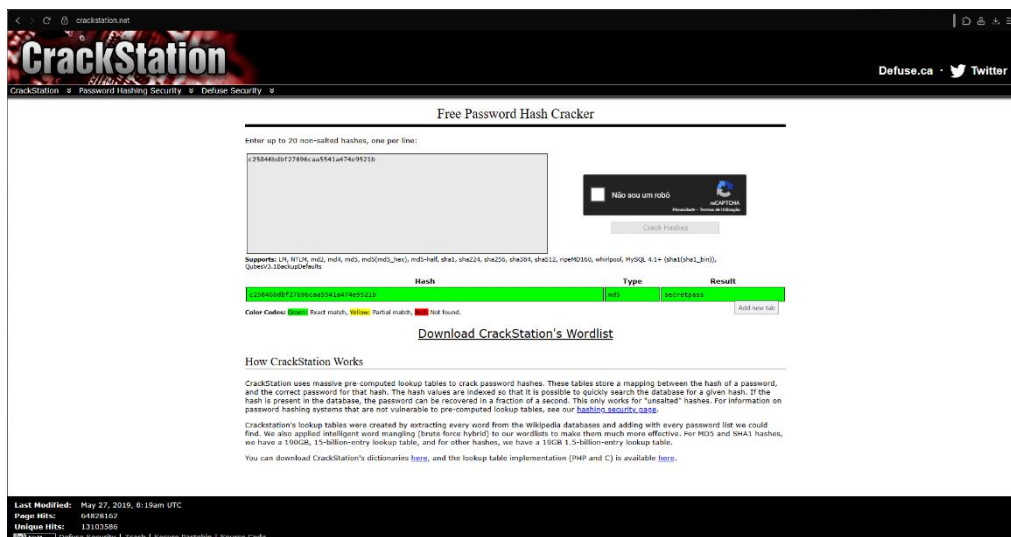
m) Foi gerada uma hash MD5 com o site proposto. De seguida, tive de recorrer à plataforma “CrackStation” visto que a plataforma proposta estava down, que reverteu a hash para a password original.



The screenshot shows the MD5 Hash Generator website. The main heading is "MD5 Hash Generator". Below it, there is a text input field containing "secretpass" and a "Generate" button. The output section shows the "Your String" as "secretpass", the "MD5 Hash" as "c25845b0727696ca0541a474c9521b", and the "SHA1 Hash" as "364bd2ed77a6544d3b711a03693ccadcc33c9d". There are "Copy" buttons for both hashes. A sidebar on the right lists "Related Tools" including "Secure database stor...", "Hash function tools", "Cybersecurity softwa...", "Programming langua...", "Add new tab", "arison tool", "Server management...", "Developer tools subs...", and "Secure Hash Algorith...".



The screenshot shows a Cloudflare error page with the message "Connection timed out (Error code 522)". It includes a "Visit cloudflare.com for more information." link and a timestamp "2026-03-04 17:37:34 UTC". Below the error message, there are three status indicators: "You Browser Working" (green checkmark), "Amsterdam Cloudflare Working" (green checkmark), and "www.md5online.org Host Error" (red X). A "What happened?" section explains that the initial connection between Cloudflare's network and the origin web server timed out. A "What can I do?" section provides advice for visitors and website owners.



The screenshot shows the CrackStation website, a "Free Password Hash Cracker". It features a text input field for a hash, a "Crack Passwords" button, and a CAPTCHA. Below the input field, there is a table with columns "Hash", "Type", and "Result". The table shows a single entry: "c25845b0727696ca0541a474c9521b" with "md5" as the type and "secretpass" as the result. A "Download CrackStation's Wordlist" link is provided. The footer includes the text "Last Modified: May 27, 2019, 8:19am UTC", "Page Hits: 64528152", "Unique Hits: 13103586", and a list of links: "Defuse Security", "Zcash", "Secure Passwords", and "Source Code".

n) Usei o comando “enable algorithm-type scrypt secret *password*”.

```
R1-0485>en
Password:
R1-0485#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1-0485 (config)#enable algorithm-type scrypt secret secretpass
R1-0485 (config)#
```

p) Esta análise revela que o protocolo Telnet transmite toda a comunicação em texto limpo, assim, conclui-se que o mesmo não é seguro para gerir sistemas remotamente.

No.	Interface name	IP Src	IP Dest	MAC Src	MAC Dest	Protocol	Info
6	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	Do Encryption Option, Will Encryption Option, Do Suppress Go Ahead, Will Terminal Type, Will Negotiate About Window Size, ...
8	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	Will Echo, Will Suppress Go Ahead, Do Terminal Type, Do Negotiate About Window Size
9	-	192.168.1.50	192.168.1.85	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	40 bytes data
10	-	192.168.1.50	192.168.1.85	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	Don't Encryption Option
12	-	192.168.1.50	192.168.1.85	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	Don't Encryption Option
13	-	192.168.1.50	192.168.1.85	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	Suboption Terminal Type
15	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	Suboption Terminal Type
16	-	192.168.1.50	192.168.1.85	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	Don't Terminal Speed
17	-	192.168.1.50	192.168.1.85	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	Do Remote Flow Control
19	-	192.168.1.50	192.168.1.85	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	Suboption Remote Flow Control
20	-	192.168.1.50	192.168.1.85	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	Don't Linemode
22	-	192.168.1.50	192.168.1.85	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	Don't New Environment Option
23	-	192.168.1.50	192.168.1.85	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	Don't Status
26	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	1 byte data
28	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	1 byte data
30	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	1 byte data
32	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	2 bytes data
34	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	1 byte data
36	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	1 byte data
38	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	2 bytes data
40	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	1 byte data
42	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	2 bytes data
43	-	192.168.1.50	192.168.1.85	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	10 bytes data
45	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	1 byte data
46	-	192.168.1.50	192.168.1.85	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	1 byte data
48	-	192.168.1.85	192.168.1.50	06:23:04:85:aa:f1	06:23:04:85:aa:f1	TELNET	1 byte data

Wireshark · Follow TCP Stream (tcp.stream eq 0) · Standard input

```

.&.&.....!.".'...
.....m....
User Access Verification
Password: .&.&.....
....XTERM..
.....!.".'...
inbandpass
.
R1-0485>
e
e
m
n
a
a
b
b
l
l
e
e
.
Password:
secretpass
.
R1-0485#
e
e
x
x
i
i
t
t
.
...!...
```

q) Configuração do router para bloquear o acesso VTY durante 90 segundos após 3 tentativas falhadas em 30 segundos através do comando “login block-for 90 attempts 3 within 30”.

```
root@A1:~# date
Wed Mar  4 18:42:23 UTC 2026
root@A1:~# telnet 192.168.1.50
Trying 192.168.1.50...
Connected to 192.168.1.50.
Escape character is '^]'.

User Access Verification

Password:
Password:
Password:
% Bad passwords
Connection closed by foreign host.
root@A1:~# date
Wed Mar  4 18:42:32 UTC 2026
root@A1:~# date
Wed Mar  4 18:45:02 UTC 2026
root@A1:~# telnet 192.168.1.50
Trying 192.168.1.50...
Connected to 192.168.1.50.
Escape character is '^]'.

User Access Verification

Password:
R1-0485>
```

r) Ativada a monitorização de autenticações no R1 através dos comandos “login on-success log” e “login on-failure log”. Foi depois confirmada a monitorização de ataques com o comando “show login”.

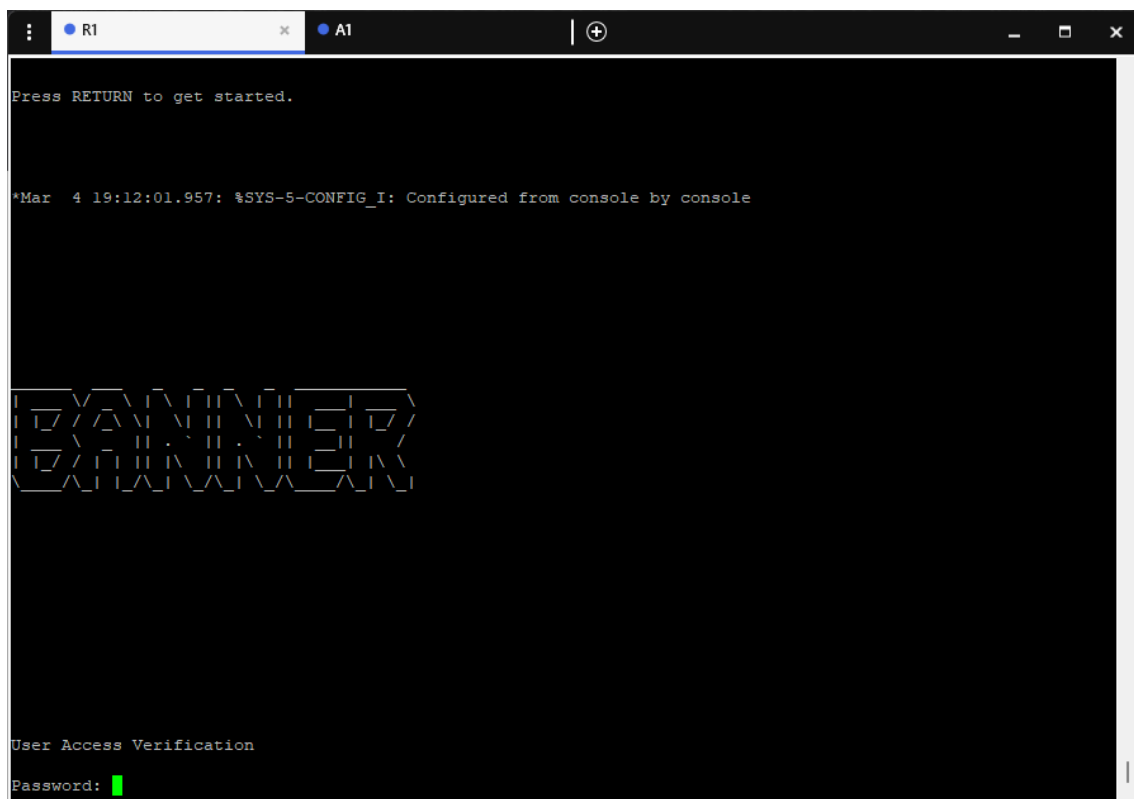
```
R1-0485#
*Mar  4 18:56:53.018: %SYS-5-CONFIG_I: Configured from console by console
R1-0485#
*Mar  4 18:57:15.029: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: admin] [Source: 192.168.1.85] [localport: 23] at 18:57:15 UTC Wed Mar 4 2026
R1-0485#
*Mar  4 18:57:18.057: %SYS-6-LOGOUT: User admin has exited tty session 2(192.168.1.85)
R1-0485#
```

```
R1-0485#show login
  A default login delay of 1 seconds is applied.
  No Quiet-Mode access list has been configured.
  All successful login is logged.
  All failed login is logged.

  Router enabled to watch for login Attacks.
  If more than 3 login failures occur in 30 seconds or less,
  logins will be disabled for 90 seconds.

  Router presently in Normal-Mode.
  Current Watch Window
    Time remaining: 28 seconds.
    Login failures for current window: 0.
  Total login failures: 0.
```

s) Definido um banner de entrada com arte gráfica.



```
Press RETURN to get started.

*Mar  4 19:12:01.957: %SYS-5-CONFIG_I: Configured from console by console

  _____
 /  _  _  _  \
|  _ \| | | | | | |
| |_) | | | | |
|  _ \| | | | |
|_| \_|_|_|_|_|

User Access Verification

Password: █
```